Environment and Health

Goal: Understand the role of environment in our health

Specific objectives:

- Describe environmental factors that affect our health
- Describe the relationship of man with the environment

Environment – is anything that externally surrounds a man. This is where man lives. The World Health Organization (WHO) defines environment, as it relates to health, as "all the physical, chemical, and biological factors external to a person, and all the related behaviors."

Environmental health – is the relationship between environment and health

Belief – is an acceptance that a statement is true or that something exists. It can also be defined as an acceptance that something exists or is true, especially one without proof

Culture - Culture refers to the cumulative deposit of knowledge, experience, beliefs, values, attitudes, meanings, hierarchies, religion, notions of time, roles, spatial relations, concepts of the universe, and material objects and possessions acquired by a group of people in the course of generations through individual and group striving

Environmental components – these are what makes up the environment. These include:

- PHYSICAL: - e.g. air, water, soil, housing, climate, geography, heat, light, noise, debris, radiation
- BIOLOGICAL – e.g. Man, viruses, microbial agents, insects, rodents, animals and plants, etc.
- PSYCHOSOCIAL – e.g. Cultural values, customs, beliefs, habits, attitudes, morals, religion, education, lifestyles, community life, health services, social and political organization.

Importance of environmental health

- Goal of environmental health is to promote health for all through a healthy environment.
- Environmental health consists of preventing or controlling disease, injury, and disability related to the interactions between people and their environment.
- It increasing quality of life and years of healthy life
- Globally, nearly 25 percent of all deaths and the total disease burden can be attributed to environmental factors such as:
    - Exposure to hazardous substances in the air, water, soil, and food
    - Natural and technological disasters
    - Emergencies
    - Water supply
    - Increased urban development
    - Physical hazards
    - Nutritional deficiencies
- Poor environmental quality has its greatest impact on people whose health status is already at risk.
- Therefore, environmental health must address the societal and environmental factors that increase the likelihood of exposure and disease.

Environmental health factors (risks) affecting our health

- Environmental health risks are factors outside of the body that can affect a person's wellbeing.
- Examples include:
    - Air pollution – for example, smog, wood smoke and mould
    - Water quality – for example, grey water, tank water, fluoridation and drought.
    - Food quality and safety – for example, contamination and nutrition.
    - Chemicals – for example, pesticides, farm chemicals.
    - Metals – for example, exposure to lead, mercury
    - Diseases from animals and insects (vector borne) – for example, dengue fever, virus, malaria.
    - Infectious diseases – for example, viral infections like swine flu.
    - Natural hazards – for example, solar radiation and extreme weather events.
    - Man-made structures – for example, exposure to asbestos or electromagnetic radiation sources like mobile phones.
    - Occupational health – for example, safety issues relating to the workplace such as noise pollution and hazardous waste.
    - Climate change – for example, higher sea levels, increased soil salinity and increased temperatures.
    - Housing conditions
- Preventing or reducing the risk of illness, injury or disease in the community is important for environmental health.

Understanding Environmental Health

- The following are 6 themes that draw attention to elements of the environment and their linkages to health.

Outdoor Air Quality

- Poor air quality is linked to premature death, cancer, and long-term damage to respiratory and cardiovascular systems.
- Decreasing air pollution is an important step in creating a healthy environment.

Surface and Ground Water

- Surface and ground water quality applies to both drinking water and recreational waters.
- Contamination by infectious agents or chemicals can cause mild to severe illness.
- Protecting water sources and minimizing exposure to contaminated water sources are important parts of environmental health.

Toxic Substances and Hazardous Wastes

- The health effects of toxic substances and hazardous wastes are not yet fully understood. Research to better understand how these exposures may impact health is ongoing.
- Meanwhile, efforts to reduce exposures continue.
- Reducing exposure to toxic substances and hazardous wastes

Homes and Communities

- People spend most of their time at home, work, or school. Some of these environments may expose people to:

- Indoor air pollution
- Inadequate heating and sanitation
- Structural problems
- Electrical and fire hazards
- Lead-based paint hazards

Infrastructure and Surveillance

- Prevention of exposure to environmental hazards relies on many partners, including state and local health departments.
- Personnel, surveillance systems, and education are important resources for investigating and responding to disease, monitoring for hazards, and educating the public.
- Additional methods and greater capacity to measure and respond to environmental hazards are needed.

Global Environmental Health

- Water quality is an important global challenge.
- Diseases can be reduced by improving water quality and sanitation and increasing access to adequate water and sanitation facilities.

Relationship of individual, family, community and environment

- Humans interact with the environment constantly. These interactions affect quality of life, years of healthy life lived, and health disparities.
- Individuals group together to make a family. Families group together to make a community. Individuals, families and communities live in the environment
- Environment hosts the individual and members of the family and community

Responsibilities if health teams in environmental health

Physician

- Performing complete health assessments including: Taking a full medical history including presenting
- Complaint, past illnesses, social history, family history, and performing a complete physical examination
- Screening patients at risk for hereditable conditions and potentially preventable disorders
- Assessment, diagnosis, primary medical treatment and advice for management of acute medical conditions and injuries
- Assessment of the exacerbations and complications of chronic medical problems

Nurses

- Assess holistically and provide services to patients in all developmental stages, and to families and communities
- Complete health assessments, including a health history and physical examination
- Formulate and communicate medical diagnoses
- Synthesize information from patients to identify broader implications for health within the family
- Use family assessment tools to evaluate family strengths and needs
- Determine the need for, and order from, an approved list of screening and diagnostic laboratory tests and interpret the results
- Determine the need for, and order and interpret reports of X-rays, ECGs and diagnostic

ultrasounds for diagnosis

Social worker

- Assessment of social problems by obtaining case history and background information
- Assessment of eligibility of social benefits

****Brainstorm roles of the following*****

- Health records and information officer
- Physiotherapy
- Occupational therapy
- Nutrition Officer
- Environmental health officer

How Culture Influences Health -Key points

- Culture is a pattern of ideas, customs and behaviors shared by a particular people or society. It is constantly evolving.
- The speed of cultural evolution varies. It increases when a group migrates to and incorporates components of a new culture into their culture of origin.
- Children often struggle with being 'between cultures'– balancing the 'old' and the 'new'. They essentially belong to both, whereas their parents often belong predominantly to the 'old' culture.
- The influence of culture on health is vast:
    - It affects perceptions of health, illness and death,
    - Beliefs about causes of disease
    - Approaches to health promotion
    - How illness and pain are experienced and expressed, where patients seek help
    - The types of treatment patients prefer.
- Both health professionals and patients are influenced by their respective cultures.
- Cultural bias may result in very different health-related preferences and perceptions.
- Demonstrating awareness of a patient's culture can promote trust, better health care, lead to higher rates of acceptance of diagnoses and improve treatment adherence.
- Acceptance of a diagnosis, including who should be told, when and how.
- Acceptance of preventive or health promotion measures (e.g., vaccines, prenatal care, birth control, screening tests, etc.).
- Perceptions of death, dying and who should be involved
- Willingness to discuss symptoms with a health care provider, or with an interpreter being present.
- Influence of family dynamics, including traditional gender roles, filial responsibilities, and patterns of support among family members.
- How accessible the health system is, as well as how well it functions.

How man has interfered with the environment

- Technology
- Agricultural land los and soil erosion
- Energy industry pollution
- Creation of infrastructure
- Coal mining and burning

- **Pharmaceuticals and personal care products**
- Creation of artificial ecosystem
- Nuclear power

Role of community in preserving Healthy and Safe Environments

What can state, local and territorial governments do?

- Facilitate collaboration among diverse sectors (e.g., planning, housing, transportation, energy, education, environmental regulation, agriculture, business associations, labor organizations, health and public health) when making decisions likely to have a significant effect on health.
- Include health criteria as a component of decision making (e.g., policy making, land use and transportation planning).
- Conduct comprehensive community health needs assessments and develop state and community health improvement plans.
- Strengthen and enforce housing and sanitary code requirements and ensure rapid remediation or alternative housing options.
- Participate in national voluntary accreditation of health departments.

What Can Businesses and Employers Do?

- Ensure that homes and workplaces are healthy, including eliminating safety hazards (e.g., trip hazards, unsafe stairs), ensuring that buildings are free of water intrusion, indoor environmental pollutants (e.g., radon, mold, tobacco smoke), and pests, and performing regular maintenance of heating and cooling systems.
- Adopt practices to increase physical activity and reduce pollution.

What Can Health Care Systems, Insurers, and Clinicians Do?

- Partner with state, local, and territorial governments, business leaders, and community-based organizations to conduct comprehensive community health needs assessments and develop community health improvement plans.
- Support integration of prevention and public health skills into health care professional training and cross train health care practitioners to implement prevention strategies.
- Increase the use of certified equipment and skilled personnel

What Can Early Learning Centers, Schools, Colleges, and Universities Do?

- Integrate appropriate core public health competencies into relevant curricula (e.g., nursing, medicine, dentistry, allied health, pharmacy, social work, education) and train professionals to collaborate across sectors to promote health and wellness.
- Implement policies and practices that promote healthy and safe environments (e.g., improving indoor air quality; addressing mold problems; reducing exposure to pesticides and lead; ensuring that drinking water sources are free from bacteria and other toxins; implementing and enforcing tobacco free policies).

What Can Community, Non-Profit, and Faith-Based Organizations Do?

- Convene diverse partners and promote strong cross-sector participation in planning, implementing, and evaluating community health efforts.

- Implement processes to ensure that people are actively engaged in decisions that affect health.

What Can Individuals and Families Do?

- Use alternative transportation (e.g., biking, walking, public transportation, car and vanpooling).
- Conduct home assessments and modifications (e.g., installing smoke and carbon monoxide detectors, testing for lead, checking for mold and radon).
- Purchase energy-efficient products, support local vendors, and recycle.

Common barriers to inter-professional healthcare team work

Organizational barriers:

- Lack of knowledge and appreciation of the roles of other health professionals;
- Compelling arguments for team building to senior decision-makers;
- Lack of outcomes research on collaboration;
- Financial and regulatory constraints;
- Legal issues of scope of practice and liability;
- Reimbursement structures for different professions, including which services receive
- Hierarchical administrative and educational structures that discourage inter-professional collaboration.

Barriers at the team level:

- Lack of a clearly stated, shared, and measurable purpose;
- Lack of training in inter-professional collaboration;
- Role and leadership ambiguity;
- Team too large or too small;
- Team not composed of appropriate professionals;
- Lack of appropriate mechanism for timely exchange of information;
- Need for orientation for new members;
- Lack of framework for problem discovery and resolution;
- Difference in levels of authority, power, expertise, income;
- Difficulty in engaging the community;
- Traditions/professional cultures, particularly medicine's history of hierarchy;
- Lack of commitment of team members;
- Different goals of individual team members;
- Apathy of team members;
- Inadequate decision making; and
- Conflict regarding individual relationships to the patient/client.

Barriers faced by individual team members:

- Split loyalties between team and own discipline;
- Multiple responsibilities and job titles;
- Competition
- Gender, race, or class-based prejudice;
- Persistence of a defensive attitude;
- Reluctance to accept suggestions from team members representing other professions; and
- Lack of trust in the collaborative process.

- Barriers for independent providers:
- Accustomed to assuming total responsibility;
- Unease with allowing others to be involved in clinical decision-making;
- Discomfort with performance review by team members of different professional backgrounds;
- Legal liability for others' decisions; and
- Dilution of traditional one-to-one relationship with patient/client.

Overcoming barriers

- Agree on unifying philosophy centered around primary care of the patient/client and the community;
- Develop a commitment to the common goal of collaboration;
- Learn about other professions;
- Respect others' skills and knowledge;
- Establish positive attitudes about own profession;
- Develop trust between members;
- Be willing to share responsibility for patient/client care;
- Establish a mechanism for negotiation and re-negotiation of goals and roles over time;
- Establish method for resolving conflicts between team members; and
- Be willing to work continuously on overcoming barriers.